First Law of Thermodynamic (Supplementary Problems)

1. In a certain process, 8.00 kcal of heat is furnished to the system while the system does 6.00 kJ of work. By how much does the internal energy of the system change during the process?

2. The specific heat of water is 4184 J/kg K. By how many joules does the internal energy of 50 g of water change as it is heated from 21 °C to 37 °C?

3. How much does the internal energy of 5.0 g of ice at precisely 0 °C increase as it is changed to water at 0 °C? Neglect the change in volume.

4. In each of the following situations, find the change in internal energy of the system. (a) A system absorbs 500 cal of heat and at the same time does 400 J of work. (b) A system absorbs 300 cal and at the same time 420 J of work is done on it. (c) Twelve hundred calories is removed from a gas held at constant volume. Give your answers in kilojoules.

5. For each of the following adiabatic processes, find the change in internal energy. (a) A gas does 5 J of work while expanding adiabatically. (b) During an adiabatic compression, 80 J of work is done on a gas.

6. How much work is done by an ideal gas in expanding isothermally from an initial volume of 3.00 liters at 20.0 atm to a final volume of 24.0 liters?

7. The P−V diagram in Fig. 20-1 applies to a gas undergoing a cyclic change in a piston−cylinder arrangement. What is the work done by the gas in (a) portion AB of the cycle? (b) Portion BC? (c) Portion CD? (d) Portion DA?

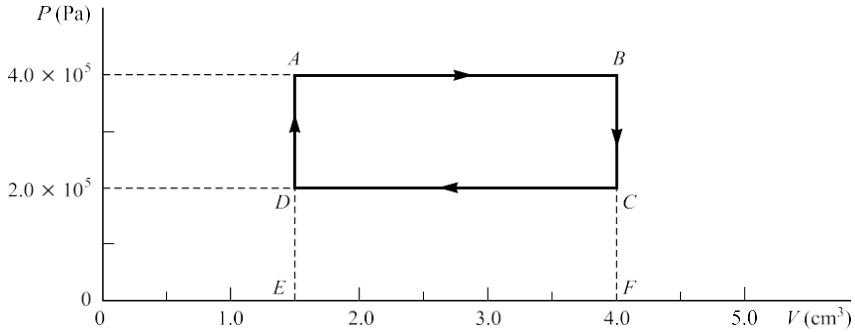


Fig. 20-1

8. For the thermodynamic cycle shown in Fig. 20-1, find (a) the net work output of the gas during the cycle and (b) the net heat flow into the gas per cycle.

9. Twenty cubic centimeters of monatomic gas at 12 °C and 100 kPa is suddenly (and adiabatically) compressed to 0.50 cm^3. What are its new pressure and temperature?

10. Compute the maximum possible efficiency of a heat engine operating between the temperature limits of 100 °C and 400 °C.

11. A steam engine operating between a boiler temperature of 220 °C and a condenser temperature of 35.0 °C delivers 8.00 hp. If its efficiency is 30.0 percent of that for a Carnot engine operating between these temperature limits, how many calories are absorbed each second by the boiler? How many calories are exhausted to the condenser each second?

12. Three kilomoles (6.00 kg) of hydrogen gas at S.T.P. expands isobarically to precisely twice its volume. (a) What is the final temperature of the gas? (b) What is the expansion work done by the gas? (c) By how much does the internal energy of the gas change? (d) How much heat enters the gas during the expansion? For H_2, c_n = 10.0 kJ/kg· K.

13. A cylinder of ideal gas is closed by an 8.00 kg movable piston (area 60.0 cm2 as shown in Fig. 20-3. Atmospheric pressure is 100 kPa. When the gas is heated from 30.0 °C to 100.0 °C, the piston rises 20.0 cm. The piston is then fastened in place, and the gas is cooled back to 30.0 °C. Calling ΔQ1 the heat added to the gas in the heating process, and ΔQ2 the heat lost during cooling, find the difference between ΔQ1 and ΔQ2.

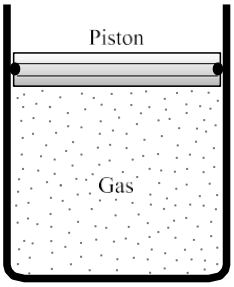


Fig. 20-3

14. A 2.0 kg metal block c = 0.137 cal/g °C is heated from 15 °C to 90 °C. By how much does its internal energy change?
Ans. 86 kJ

15. By how much does the internal energy of 50 g of oil (c = 0.32 cal/g·°C) change as the oil is cooled from 100 °C to 25 °C.
Ans. -1.2 kcal

16. How many joules of heat per hour are produced in a motor that is 75.0 percent efficient and requires 0.250 hp to run it? Ans. 168 kJ

17. How much external work is done by an ideal gas in expanding from a volume of 3.0 liters to a volume of 30.0 liters against a constant pressure of 2.0 atm? Ans. 5.5 kJ

18. As 3.0 liters of ideal gas at 27 °C is heated, it expands at a constant pressure of 2.0 atm. How much work is done by the gas as its temperature is changed from 27 °C to 227 °C? Ans. 0.40 kJ

19. An ideal gas expands adiabatically to three times its original volume. In doing so, the gas does 720 J of work. (a) How much heat flows from the gas? (b) What is the change in internal energy of the gas? (c) Does its temperature rise or fall? Ans. (a) none; (b) —720 J; (c) it falls

20. An ideal gas expands at a constant pressure of 240 cmHg from 250 cm^3 to 780 cm^3. It it then allowed to cool at constant volume to its original temperature. What is the net amount of heat that flows into the gas during the entire process? Ans. 40.4 cal

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22. As an ideal gas is compressed isothermally, the compressing agent does 36 J of work. How much heat flows from the gas during the compression process? Ans. 8.6 cal

23. The specific heat of air at constant volume is 0.175 cal/g ·°C. (a) By how much does the internal energy of 5.0 g of air change as it is heated from 20 °C to 400 °C? (b) Suppose that 5.0 g of air is adiabatically compressed so as to rise its temperature from 20 °C to 400 °C. How much work must be done on the air to compress it? Ans. (a) 0.33 kcal; (b) 1.4 kJ or since work done on the system is negative, -1.4 kJ

24. Compute the work done in an isothermal compression of 30 liters of ideal gas at 1.0 atm to a volume of 3.0 liters. Ans. 7.0 kJ

25. Find the net work output per cycle for the thermodynamic cycle in Fig. 20-4. Give your answer to two significant figures. Ans. 2.1 kJ

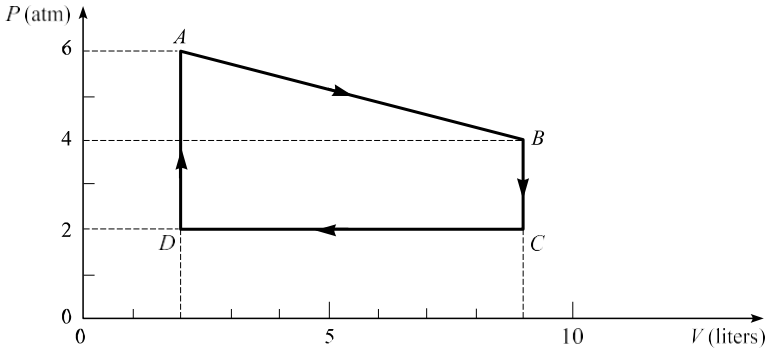


Fig. 20-4

26. Four grams of gas, confined to a cylinder, is carried through the cycle shown in Fig. 20-4. At A the temperature of the gas is 400 °C. (a) What is its temperature at B? (b) If, in the portion from A to B, 2.20 kcal flows into the gas, what is c_n for the gas? Give your answers to two significant figures. Ans. (a) 2.0 × 103 K; (b) 0.25 cal/g ·°C

27. Figure 20-4 is the P−V diagram for 25.0 g of an enclosed ideal gas. At A the gas temperature is 200 °C. The value of cn for the gas is 0.150 cal/g °C. (a) What is the temperature of the gas at B? (b) Find ΔU for the portion of the cycle from A to B. (c) Find ΔK for this same portion. (d) Find ΔQ for this same portion. Ans. (a) 1.42 × 103 K; (b) 3.55 kcal = 14.9 kJ; (c) 3.54 kJ; (d) 18.4 kJ

28. Determine values for the work done in the two changes of state represented in Figure B4.6.

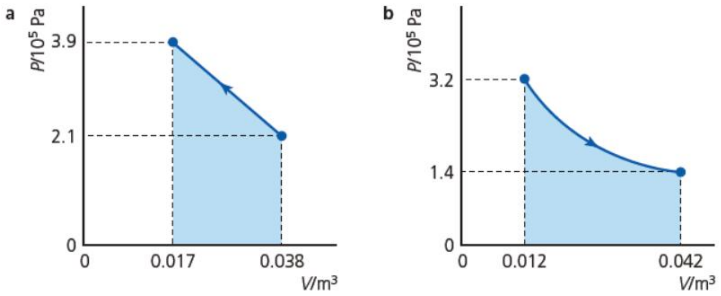


Figure B4.6 Determining areas under pressure–volume graphs

29. The volume of a gas expanded from 43 cm^3 to 51 cm^3 while its pressure remained constant at 1.1×10^5 Pa.

a Calculate the work done.

b State whether work was done on the gas, or by the gas.

30. 0.84 J of work was done on a gas of volume 7.6×10^{-5} m^3. If the pressure was constant at 1.4 × 105 Pa, calculate the new volume.

31. Figure B4.8 shows how the volume of a gas changed as the pressure on it was increased. Show that the work done during the expansion was approximately 230 J.

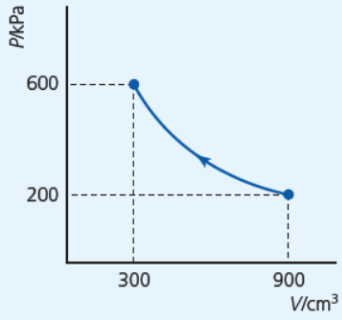


Figure B4.8 *PV* graph for a gas

32. 4 200 J of work was done when a gas expanded at constant temperature. How much thermal energy was transferred?

33. 70 J of work was done while a gas was compressed. At the same time 30 J of thermal energy was transferred out of the gas. Calculate the change in internal energy.

34. 50 J of thermal energy was removed from a gas and its internal energy decreased by 10 J.
a How much work was done?
b Was the work done on the gas, or by the gas?

35. An ideal monatomic gas containing 3.4×10^{24} particles was heated from 297 K to 348 K.

a What was the change in internal energy of the gas?

b If, in this time, 5600 J of thermal energy was supplied to the gas, what was the amount of work done by, or on, the gas?

36. The internal energy of 3.2 moles of some helium gas, at an initial temperature of 302 K, rose by 540 J.

a What was the final temperature of the gas?

b If, at the same time, the gas expanded from a volume of 220 cm^3 to 380 cm^3 at constant pressure of 2.3×10^5 Pa, how much work was done?

c How much thermal energy was transferred?

37. During an investigation of how the volume of a fixed mass of air changed with temperature at constant atmospheric pressure (1.0×10^5 Pa), the volume increased from 2.3 cm^3 to 3.1 cm^3.
a Calculate how much work was done.
b Was the work done on the gas, or by the gas?
c The expansion of the air happened when 0.150 J of thermal energy was transferred into the system. Determine the change in internal energy of the gas.

38. 1.82 J of thermal energy was supplied to a gas, increasing its pressure steadily from 1.0×10^5 Pa to 1.2×10^5 Pa. During this time the volume was increased steadily from 35 cm^3 to 45 cm^3. a Represent this process on a PV diagram. b Indicate the work done during this process on your drawing. c Calculate the work done. d Determine the change in internal energy of the gas.

39. Consider Figure B4.9. Work has been done to change the state of a gas from point A to point B on the PV diagram.

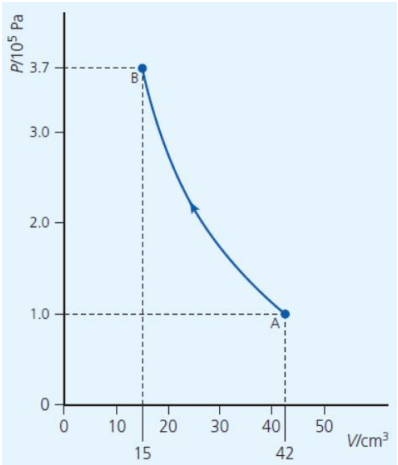


Figure B4.9 *PV* graph

a Describe the process.
b Did the process obey Boyle's law?
c What has happened to the temperature of the gas? Explain your answer.
d Estimate the work done during the process.
e During the process, 1.9 J of thermal energy flowed out of the gas. Determine the change in its internal energy.

40. 1.74 mol of an ideal monatomic gas expanded and during the process its temperature changed. The container was well insulated and no thermal energy was able to flow into, or out of, the gas.

a If the internal energy of the gas decreased by 29.7 J, calculate the temperature change of the gas.

b Determine how much work was done by the gas during this expansion.

41. A gas of volume 0.080 m^3 and pressure 1.4×10^5 Pa expands to a volume of 0.11 m^3 at constant pressure when 7.4×10^3 J of thermal energy are supplied.

a Name the thermodynamic process.

b Calculate the work done by the gas.

c Determine the change in the internal energy of the gas.

42. What is the total internal energy of a monatomic ideal gas that contains 4.0 moles at a temperature of 400 K?

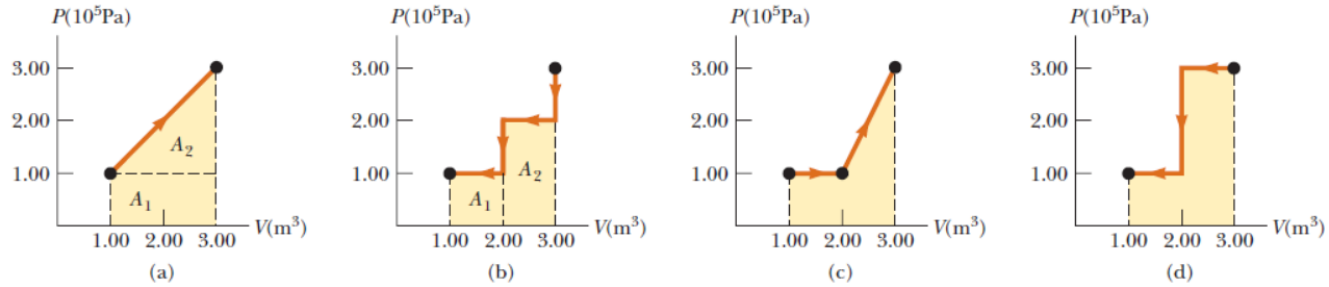
43. A system takes in 5000 J of heat and does 4000 J of work on its surroundings. What is the change in the internal energy of the system?

44. SSM The temperature of an ideal monatomic gas increases by 100 K while 3000 J of heat are added to the gas. If the gas contains 4.0 moles, how much work does it do on its surroundings?

45. The temperature of a sample of dilute argon gas with n = 7.5 moles decreases by 200 K. If 35,000 J of heat are extracted from the gas, what is the work done by the gas on its surroundings?

46. One gallon of gasoline releases approximately 1.2×10^8 J when it is "burned" (i.e., undergoes combustion) in a car engine. If the engine releases 4.0×10^7 J of heat during this time, what is the mechanical work done by the engine?

47. Find the numeric value of the work done on the gas in (a) Figure 12.3a, 12.3b, 12.3c and 12.3d.



48. (a) Determine the work done on a fluid that expands from i to f as indicated in Figure P12.10. (b) How much work is done on the fluid if it is compressed from f to i along the same path?

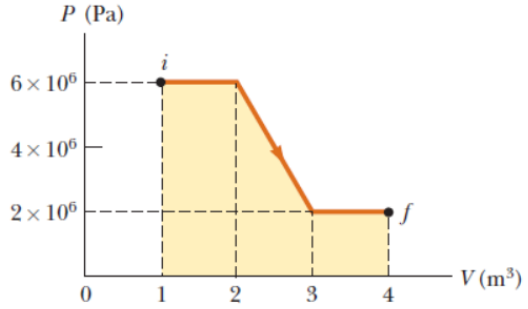


Figure P12.10

49. A container is placed in a water bath and held at constant volume as a mixture of fuel and oxygen is burned inside it. The temperature of the water is observed to rise during the burning. (The water is also held at constant volume.

(a) Consider the burning mixture to be the system. What are the signs of Q, ΔU, and W?

(b) What are the signs of these quantities if the water bath is considered to be the system?

50. A gas is compressed at a constant pressure of 0.800 atm from 9.00 L to 2.00 L. In the process, 400 J of energy leaves the gas by heat. (a) What is the work done on the gas? (b) What is the change in its internal energy?

51. A quantity of a monatomic ideal gas undergoes a process in which both its pressure and volume are doubled as shown in Figure P12.12. What is the energy absorbed by heat into the gas during this process?

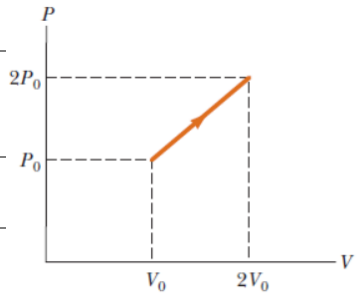


Figure P12.12

52. A monatomic ideal gas undergoes the thermodynamic process shown in the PV diagram of Figure P12.14. Determine whether each of the values ΔU, Q, and W for the gas is positive, negative, or zero.

53. A gas expands from I to F in Figure P12.5. The energy added to the gas by heat is 418 J when the gas goes from I to F along the diagonal path. (a) What is the change in internal energy of the gas? (b) How much energy must be added to the gas by heat for the indirect path IAF to give the same change in internal energy?

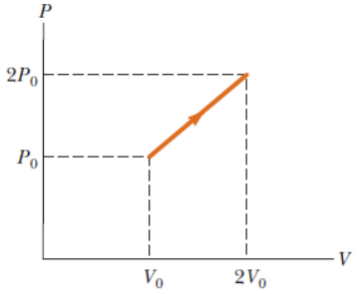


Figure P12.12

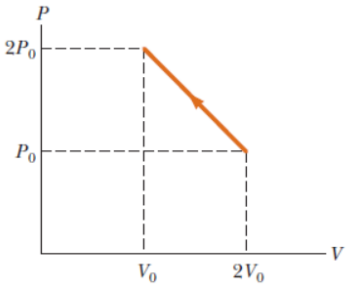


Figure P12.14

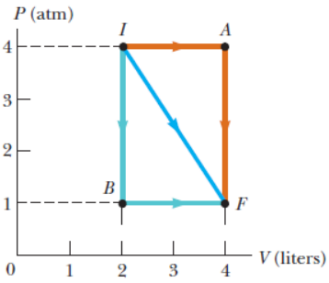


Figure P12.5 (Problems 5 and 15)

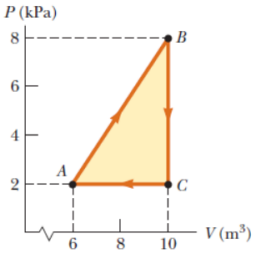


Figure P12.16 (Problems 16 and 18)

54. A gas is taken through the cyclic process described by Figure P12.16.
(a) Find the net energy transferred to the system by heat during one complete cycle.
(b) If the cycle is reversed—that is, the process follows the path ACBA— what is the net energy transferred by heat per cycle?

55. A gas is enclosed in a container fitted with a piston of cross-sectional area 0.150 m2. The pressure of the gas is maintained at 6 000 Pa as the piston moves inward 20.0 cm. (a) Calculate the work done by the gas. (b) If the internal energy of the gas decreases by 8.00 J, find the amount of heat removed from the system by heat during the compression.

56. Consider the cyclic process described by Figure P12.16. If Q is negative for the process BC and ΔU is negative for the process CA, determine the signs of Q, W, and ΔU associated with each process.

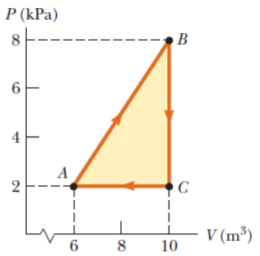


Figure P12.16 (Problems 16 and 18)

57. One gram of water changes to ice at a constant pressure of 1.00 atm and a constant temperature of 0 °C. In the process, the volume changes from 1.00 cm^3 to 1.09 cm^3. (a) Find the work done on the water and (b) the change in the internal energy of the water.

58. A thermodynamic system undergoes a process in which its internal energy decreases by 500 J. If at the same time 220 J of work is done on the system, find the energy transferred to or from it by heat. A 5.0-kg block of aluminum is heated from 20 C to 90 C at atmospheric pressure. Find
(a) the work done by the aluminum,
(b) the amount of energy transferred to it by heat, and
(c) the increase in its internal energy.

59. One mole of gas initially at a pressure of 2.00 atm and a volume of 0.300 L has an internal energy equal to 91.0 J. In its final state, the gas is at a pressure of 1.50 atm and a volume of 0.800 L, and its internal energy equals 180 J. For the paths IAF, IBF, and IF in Figure P12.22, calculate

(a) the work done on the gas and

(b) the net energy transferred to the gas by heat in the process.

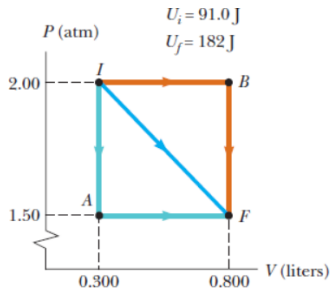


Figure P12.22